

Chemistry Exam 13 — Chapters 1-9 (Full Syllabus)

Seemab Ali | Class 9 | Federal Board (FBISE) | Version 2 | Attempt date: 2026-06-30 | Marked: 2026-06-30

CORRECTED 2026-06-30: Q3(iii)(b) was not attempted (0/2); score revised from 62.5/65 to 60.5/65.

CORRECTION: The original report awarded 5/5 for Q3(iii) and described S/N oxidation-state working that does not exist on the scan. On re-examination at 300 DPI, the answer sheet shows only the three redox definitions (part a, 3/3). Part (b) — oxidation states of S in H₂SO₄ and N in HNO₃ — is entirely absent; the writing runs straight from the redox definitions into Q3(iv). Corrected Q3(iii) = 3/5. Total: 60.5/65 (93.1%, A+).

12/12 Section A (MCQs)	31.5/33 Section B (Short)	17/20 Section C (Long)	60.5/65 Total — 93.1%	A+ Grade
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Section A — Multiple Choice Questions (12/12 — Perfect Score)

Q	Question Summary	Correct	Seemab	Result	Marks
1	Branch studying physical/chemical changes in living organisms	B	B	✓	1
2	Electrically conductive state found in sun and lightning	C	C	✓	1
3	Major defect of Rutherford's atomic model	B	B	✓	1
4	Neutrons in atom with Z=17, A=35	C (18)	C	✓	1
5	Highest electronegativity on Pauling scale	C (F)	C	✓	1
6	Halogen that is a red-brown liquid at room temperature	C (Bromine)	C	✓	1
7	In coordinate covalent bond, both electrons donated by	C (one atom — donor)	C	✓	1
8	Formula mass of KCl	B (74.5 amu)	B	✓	1
9	Molecules in 0.5 mol of CH ₄	B (3.011×10 ²³)	B	✓	1
10	Oxidizing agent in 2Na + Cl ₂ → 2NaCl	B (Cl ₂)	B	✓	1
11	How a catalyst increases reaction rate	B (Lowers activation energy)	B	✓	1
12	True statement about chemical equilibrium	D (Both reactions continue at same rate)	D	✓	1

Perfect MCQ score: 12/12 (100%). This is Seemab's 9th consecutive perfect MCQ score across all subjects — a new streak record. All 12 answers correct first attempt, covering all 9 chapters.

Section B — Short Questions (31.5/33)

Q2(i) — MAIN

Industrial chemistry vs nuclear chemistry; one example each

3/3

Both definitions correct and well phrased. Nuclear chemistry: studies changes in nuclei of atoms including radioactive decay; example Uranium-238. Industrial chemistry: large-scale production of chemical substances for mankind; example ammonia by the Haber process. Full marks awarded.

Clear two-column layout makes the differentiation immediately readable.

Q2(ii) — MAIN

Macroscopic properties of solids, liquids, gases in tabular form

3/3

Three properties tabulated (density, compressibility, ability to flow) across all three states. Solid: high density, not compressible, cannot flow. Liquid: high density, moderate compressibility, can flow. Gas: low density, compressibility "high", ability to flow "Yes". The gas column uses informal shorthand but the factual content is correct. Full marks awarded.

Q2(iii) — OR

Isotopes: define, name three H isotopes with mass numbers, identify radioactive one

3/3

Isotopes correctly defined as atoms of same element with same atomic number but different mass numbers. Table provided: Protium (H¹, mass 1, neutrons 0), Deuterium (H², mass 2, neutrons 1), Tritium (H³, mass 3, neutrons 2) — all correct. Tritium (H³) correctly identified as radioactive. Full marks.

Q2(iv) — MAIN

Electronic configuration of Na (Z=11) and Al (Z=13); group and period

2.5/3

Na: 1s²2s²2p⁶3s¹, Group 1, Period 3rd — all correct (1.5/1.5). Al: 1s²2s²2p⁶3s²3p¹ — configuration correct. Period 3rd correct. Group written as "3+10=13 or 3" — the working shows Group 13 but the final stated answer vacillates between "13" and "3", creating genuine ambiguity.

−0.5: Group for Al left ambiguous ("or 3"). The correct answer is Group 13. Writing "3" alongside it introduces doubt about whether the d-block offset was understood. Configuration is correct so only 0.5 deducted rather than 1.

Q2(v) — OR Define ionic bond; derive formula of CaCl₂ from ionic charges**3/3**

Ionic bond defined precisely as "strong electrostatic force of attraction between oppositely charged ions (positively charged metal cation and negatively charged non-metal anion), forms when an electropositive metal transfers electrons to an electronegative non-metal." Deriving CaCl₂: Ca²⁺ (+2), Cl⁻ (-1) identified; compound must be electrically neutral; two Cl⁻ needed to balance +2; formula CaCl₂ derived. All three steps present. Full marks.

Q2(vi) — OR Mass of 2.5 mol sucrose C₁₂H₂₂O₁₁; number of molecules**3/3**

Molar mass: 12(12)+22(1)+11(16) = 144+22+176 = 342 g/mol (correct). Mass = 2.5 × 342 = 855 g (correct). Molecules = 2.5 × 6.022×10²³ = 1.506×10²⁴ (correct). All three marks awarded.

Q2(vii) — OR Moles in 100 g of H₂O and in 5 g of H₂**3/3**

H₂O: molar mass = 2(1)+16 = 18 g/mol; moles = 100/18 = 5.56 mol (correct). H₂: molar mass = 2(1) = 2 g/mol; moles = 5/2 = 2.5 mol (correct). Both calculations shown with working. Full marks.

Q2(viii) — OR Identify oxidizing and reducing agents in Fe₂O₃ + 3CO → 2Fe + 3CO₂**3/3**

Oxidizing agent defined (causes oxidation by accepting electrons; itself reduced) — 1mk. Reducing agent defined (causes reduction by donating electrons; itself oxidised) — 1mk. In the equation: Fe +3→0 (reduced, oxidising agent = Fe₂O₃) ✓; C +2→+4 (oxidised, reducing agent = CO) ✓ — 1mk. All correct. Full marks.

Q2(ix) — OR KMnO₄ + reducing agent: colour change and oxidation number of Mn**3/3**

Colour change: purple to colourless (correct) — 1mk. Explanation: KMnO₄ is strong oxidizing agent; MnO₄⁻ (purple) reduced to Mn²⁺ (colourless), discharging purple colour — 1mk. Oxidation number of Mn: +7 (in MnO₄⁻) to +2 (in Mn²⁺), decrease of 5, represents reduction — 1mk. Full marks.

Q2(x) — MAIN Define enthalpy of reaction; ΔH sign for exo/endothermic; one thermochemical equation each**2/3**

Enthalpy of reaction defined as "amount of heat or thermal energy evolved or absorbed when a chemical reaction takes place; denoted by ΔH" — correct (1mk). Exothermic: ΔH negative, heat released to surroundings (correct). Endothermic: ΔH positive, heat absorbed from surroundings (correct). Signs for both correctly stated.

—1: No thermochemical equations provided. The question explicitly asks for "one thermochemical equation for each" (exothermic and endothermic). Two equations with ΔH values were required and both are missing. The definitions of the sign conventions are correct but the worked examples are not present.

Q2(xi) — OR Three conditions for equilibrium; what is dynamic equilibrium?**3/3**

Three conditions: (1) concentration of reactants or products remains unchanged ✓, (2) temperature of the system remains constant ✓, (3) pressure or volume of the system remains constant ✓ — 2mks. Dynamic equilibrium: "both the forward and reverse reactions are still occurring at the same rate; the system is not static; concentrations of reactants and products remain constant because the rates of the two opposing reactions are equal" — excellent, fully correct definition covering all key aspects — 1mk. Full marks.

This is a notably strong answer — better phrased than the answer key's own wording.

Section C — Long Questions (17/20)**Q3(i) — MAIN (a) Dalton's theory: 3 postulates + 1 limitation (3mks) | (b) Z=19, A=39: subatomic particles, config, group, period (2mks)****4/5**

Part (a): Three postulates stated: (1) all matter composed of tiny indivisible particles called atoms ✓, (2) atoms of a particular element are identical in mass and properties ✓, (3) atoms can neither be created nor be destroyed ✓. That earns 2/2 for postulates.

—1: Limitation not stated. The question asks explicitly for "one limitation." The required answer is that Dalton stated atoms are indivisible, but we now know atoms consist of subatomic particles (protons, neutrons, electrons) — and his model could not explain isotopes. No limitation appears in Seemab's answer. Full 1mk lost.

Part (b): Z=19, A=39. Protons=19 ✓, electrons=19 ✓, neutrons=20 ✓. Electronic configuration: 1s²2s²2p⁶3s²3p⁶4s¹ ✓. Group 1, Period 4th ✓. 2/2.

Q3(ii) — OR (a) Empirical vs molecular formula; benzene empirical=CH, M=78 amu — find molecular formula (3mks) | (b) Atoms in 1.25 mol Zn and 0.2 mol Al (2mks)**5/5**

Part (a): Empirical formula = simplest whole number ratio of atoms in a compound ✓. Molecular formula = actual number of each type of atom in one molecule ✓. Distinction earns 1mk. Benzene: empirical formula mass = 12+1 = 13 amu ✓. n = 78/13 = 6 ✓. Molecular formula = (CH)₆ = C₆H₆ ✓. 2mks. Full 3mks for part (a). Part (b): Atoms in 1.25 mol Zn = 1.25 × 6.022×10²³ = 7.528×10²³ atoms ✓. Atoms in 0.2 mol Al = 0.2 × 6.022×10²³ = 1.2044×10²³ atoms ✓. Both calculations correct with working shown. 2mks.

Q3(iii) — MAIN (a) Redox in terms of O transfer, H transfer, electron transfer with examples (3mks) | (b) Oxidation state of S in H₂SO₄ and N in HNO₃ (2mks)**3/5****CORRECTED (was 5/5 in original report).**

Part (a): (i) Oxygen transfer: oxidation = gain of O, reduction = loss of O. Example: $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ (CO gains O = oxidised; Fe₂O₃ loses O = reduced) ✓ 1mk. (ii) Electron transfer: oxidation = loss of electrons, reduction = gain of electrons. Example: $\text{Cu}^{2+} + \text{Zn} \rightarrow \text{Cu} + \text{Zn}^{2+}$ (Zn loses 2e = oxidised, Cu²⁺ gains 2e = reduced) ✓ 1mk. (iii) Hydrogen transfer: oxidation = loss of H, reduction = gain of H. Example: $2\text{C}_2\text{H}_2 + 5\text{O}_2 \rightarrow 4\text{CO}_2 + 2\text{H}_2\text{O}$ (C₂H₂ loses hydrogen = oxidised) ✓ 1mk. Part (a) total: 3/3.

–2 (part b not attempted): The oxidation states of S in H₂SO₄ and N in HNO₃ were required. Confirmed at 300 DPI: the answer sheet runs straight from the three redox definitions into Q3(iv) — no working for part (b) appears anywhere on the page. The original report incorrectly described S/N working that does not exist on the scan. Award 0/2 for part (b). This is the single largest mark loss in this paper.

The three redox definitions in part (a) are correct with distinct examples (oxygen, electron, hydrogen transfer) — those 3 marks are fully earned.

Q3(iv) — OR (a) Effect of temperature on hydrated cobalt(II) chloride equilibrium; equation; colour changes (3mks) | (b) System and surroundings; one exo and one endothermic equation with ΔH (2mks)**5/5**

Part (a): Equation: $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ (pink solid) $\rightleftharpoons$ CoCl_2 (blue solid) + $6\text{H}_2\text{O}$ ✓ 1mk. On heating: temperature increases, equilibrium shifts to the right (forward reaction endothermic), absorbs heat to reduce the change; hydrated cobalt(II) chloride (pink) loses water and becomes anhydrous cobalt(II) chloride (blue) ✓ 1mk. Colour change pink→blue on heating correctly stated ✓. 3/3. Part (b): System = part of the universe under study in a chemical reaction ✓. Surroundings = everything else outside the system ✓. Exothermic: $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$ ΔH° = –393.5 kJ ✓. Endothermic: $\text{N}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO(g)}$ ΔH° = +180.5 kJ ✓. 2/2.

Strengths in This Attempt

- Perfect MCQ score (12/12) — 9th consecutive perfect MCQ across all subjects
- All three redox definitions (O, H, electron transfer) in Q3(iii)(a) correct with valid distinct examples — 3/3 on part (a)
- Dynamic equilibrium definition (Q2(xi)OR) was exemplary — better than the model answer
- Stoichiometry calculations all error-free: sucrose mass, molecules, moles of H₂O and H₂ (Q2(vi), Q2(vii), Q3(ii)(b))
- Cobalt(II) chloride equilibrium (Q3(iv)OR) fully addressed: equation, colour changes, system/surroundings, and both ΔH equations

Areas to Revise

- Q3(iii)(b) — Oxidation state assignment not attempted (–2 mks, largest single loss):** S in H₂SO₄ = +6; N in HNO₃ = +5. The method: assign known oxidation states (H=+1, O=–2), set sum to zero, solve for unknown. Both were required; neither appeared on the answer sheet. This is another instance of the recurring pattern of stopping before completing every required sub-part.
- Q3(i)(a) — Dalton's limitation omitted (–1 mk):** Three postulates written but the limitation was not stated. Required: atoms thought indivisible but subatomic particles (protons, neutrons, electrons) were later discovered; model also cannot explain isotopes.
- Q2(x) — No thermochemical equations written (–1 mk):** Signs for ΔH correctly defined but both required equations (one exothermic, one endothermic, with ΔH values) are missing.
- Q2(iv) — Al group ambiguous "13 or 3" (–0.5 mk):** Always write the final Group number clearly. Group 13, not "or 3".

Chemistry Attempts — Comparison

Date	Exam	MCQ	Total	%	Grade
12 Apr 2026	exam-01-ch1-2	12/12	64/65	98.5%	A+
17 Apr 2026	exam-02-ch3-5	12/12	59.5/65	91.5%	A+
12 May 2026	exam-03-ch6	12/12	57.5/65	88.5%	A+
12 Jun 2026	exam-04-ch7	12/12	60/65	92.3%	A+
24 Jun 2026	exam-12-ch1-9	12/12	62.5/65	96.2%	A+
30 Jun 2026	exam-13-ch1-9 (this exam — CORRECTED)	12/12	60.5/65	93.1%	A+

exam-13 (60.5/65, 93.1%) sits just below the other full-syllabus paper exam-12 (62.5/65, 96.2%); both are clear A+ results with perfect MCQ. Chemistry subject average is $364.0/390 = 93.3\%$ across all 6 papers. The MCQ perfect-streak now stands at 9 consecutive across all subjects.